

Abstract

In rural India, accessing retina diagnosis is challenging due to distant medical facilities and specialist shortages. Traditional diagnosis requires eye specialists to analyze fundus photographs, making the process expensive and inaccessible for villagers. With retinal diseases potentially causing irreversible blindness, and building new facilities or training more specialists being impractical, machine learning offers a viable solution. By training ML and DL models on expert-annotated fundus images, we can create systems that rapidly screen for retinal diseases, referring only complex cases to specialists, significantly reducing costs and improving healthcare access in remote communities. In this project work, we have compared machine learning algorithms like Decision Tree, Random Forest, KNN, and deep learning models like DenseNet121, ResNet101, and ResNet50 for the detection of retinal diseases that demonstrate a clear performance advantage for deep learning approaches. While the classical algorithms achieved respectable accuracy levels, peaking at 84.95% for KNN, deep learning models notably outperformed them, with ResNet50 reaching the highest overall accuracy of 96.22%. Moreover, the precision, recall, specificity, and F1 scores were consistently higher for deep learning models, indicating their superior ability to correctly identify disease cases and reduce both false positives and false negatives.

Keywords: Machine Learning (ML), Deep Learning (DL), Retinal Diseases, Medical image analysis, Feature Extraction, Retinopathy.

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34	Formula of Specificity
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36	Formula of FPR
37	Formula of FNR

List of Abbreviations/Symbols

No.	Abbreviation
1	ML - Machine Learning
2	DL - Deep Learning
3	IDRiD - Indian Diabetic Retinopathy Image Dataset
4	HRF - High-Resolution Fundus Image Database
5	KNN - K Nearest Neighbour
6	DR - Diabetic Retinopathy
7	CNN - Convolutional Neural Networks
8	EDA - Exploratory Data Analysis
9	AI - Artificial Intelligence
10	FPR - False Positive Rate
11	FNR - False Negative Rate
12	FC - Fully Connected
13	ResNet - Residual Network

1. Introduction

This project is about finding retinal diseases using Machine Learning (ML) and Deep Learning (DL) models. These computer models can help find eye problems quickly and at a lower cost. That means more people can get checked and treated on time, without spending too much money or waiting for long hours. A retinal disease is any problem or damage to the retina. The retina is a very important part of the eye. If it gets damaged and is not treated early, it can lead to serious vision loss or even blindness. If we find the problem early, doctors can help and treat the eye before it gets worse.

The retina is a thin layer at the back of the eye. It is made up of millions of tiny cells called rods and cones. These cells help us see by catching the light that comes into our eyes. The retina then sends this information to the brain through the optic nerve. This is how we see everything around us. If you look at Figure 1 below, you will see where the retina is and how the different parts of the eye are placed. By understanding the eye's structure, we can see why the retina is so important. This is why it is important to detect retinal diseases early by using Machine Learning (ML) and Deep Learning (DL). This can help many people keep their eyesight, especially in places where it is hard to visit an eye doctor.

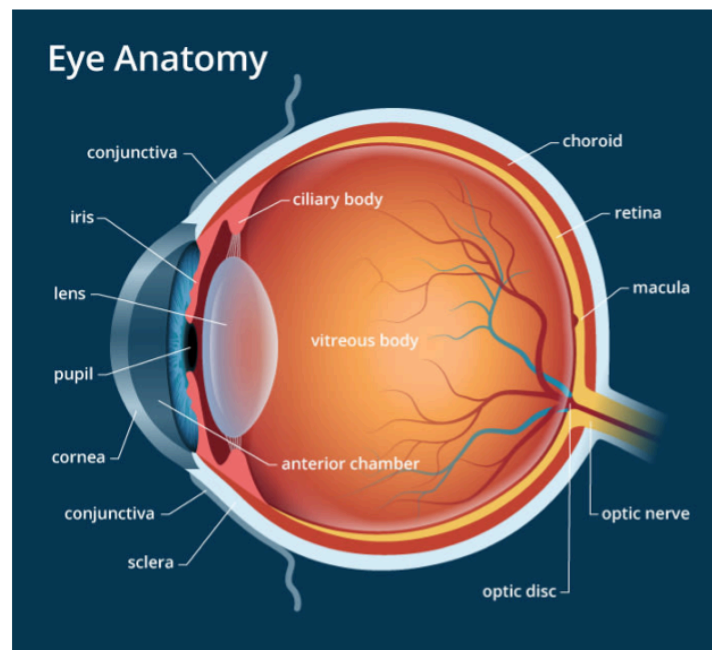


Figure 1. The structure of the Eye.

Some of the main retinal diseases we are focusing on in this project are Cataract, Diabetic Retinopathy, and Glaucoma. All these problems can make it hard to see and may even lead to blindness if they are not treated. Let's start with cataract. Cataract happens when the lens of the eye becomes cloudy. The lens is the clear part of the eye that focuses light onto the retina, so we can

see clearly. When someone has a cataract, things can look blurry or cloudy, kind of like looking through a fogged-up window. Cataracts are very common and are the number one cause of blindness in the world. Many older people get cataracts as they age, but they can happen to younger people, too.

Next is diabetic retinopathy (often called DR). As the name suggests, this disease is related to diabetes, which is a problem with high blood sugar. If a person has diabetes for many years, high blood sugar can damage the tiny blood vessels in the retina. The retina is the part of the eye that senses light and allows us to see. When these blood vessels become weak or start leaking, it can hurt the retina and cause vision problems. People with high blood pressure or high cholesterol are also at a higher risk of getting diabetic retinopathy. Over time, if left untreated, this damage can become very serious and make it hard to see.

The third disease is glaucoma. Glaucoma is caused when the pressure inside the eye gets too high, but sometimes it can even happen with normal eye pressure. This pressure can slowly damage the optic nerve, which sends signals from the eye to the brain. Glaucoma often doesn't have clear warning signs at first. There are two common types of glaucoma. One is open-angle glaucoma, which gets worse very slowly and can go unnoticed for years—most people don't feel any pain and only notice vision problems when it's already very bad. The other type is closed-angle glaucoma, which can happen suddenly. This type is painful and can cause vision loss very quickly. Both types can lead to blindness if not treated on time.

To help doctors find these diseases, we use special images called fundus images. The fundus is the inside, back surface of the eye. It includes several important parts: the retina, macula, optic disc, fovea, and a network of tiny blood vessels. Fundus photography uses a special camera, called a fundus camera, which can take detailed photos of the inside of the eye by looking through the pupil. These pictures are very helpful because they let eye doctors see if the retina or other parts look unhealthy or damaged.

In this project, we will use these fundus images as the main data to train our Machine Learning (ML) and Deep Learning (DL) models. By looking at these photos, the models learn to spot the signs of cataract, glaucoma, or diabetic retinopathy. If you look below at the figures, you can see examples of these images. There is a picture of a normal, healthy retina (Figure 2, Normal retina), then a retina with cataract (Figure 3, Retina with cataract), a retina with glaucoma (Figure 4, Retina with glaucoma), and a retina with diabetic retinopathy (Figure 5, Retina with diabetic retinopathy). By comparing these photos, it becomes easier to see what changes happen in each disease and how different a healthy eye is from a diseased one. With these clear pictures and the help of computers, it becomes much easier and faster to find out if someone has one of these serious eye problems. The sooner these problems are found, the earlier the treatment will start, and the better people's chances are to save their eyesight.

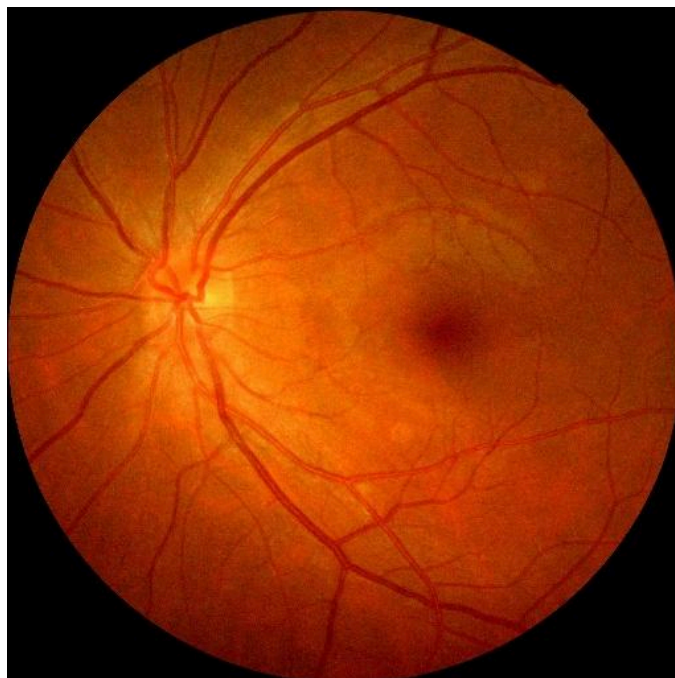


Figure 2. Normal Retina

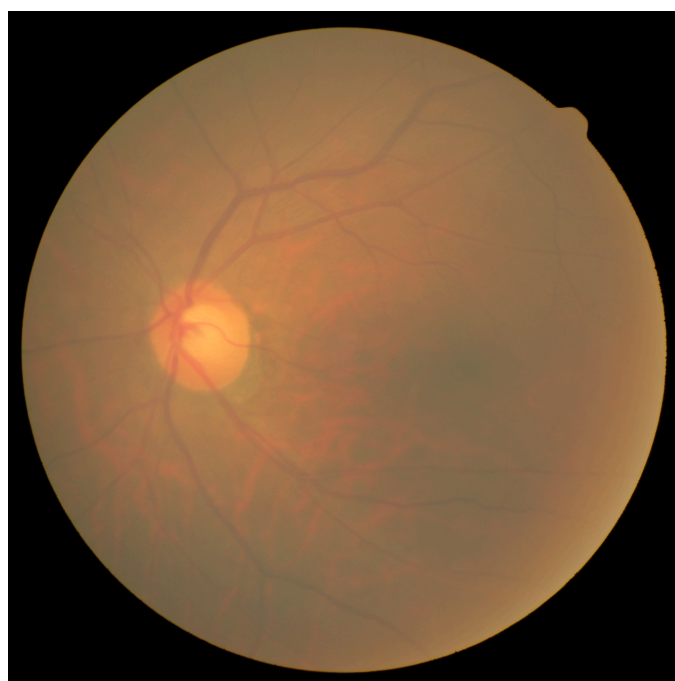


Figure 3. Retina with Cataract

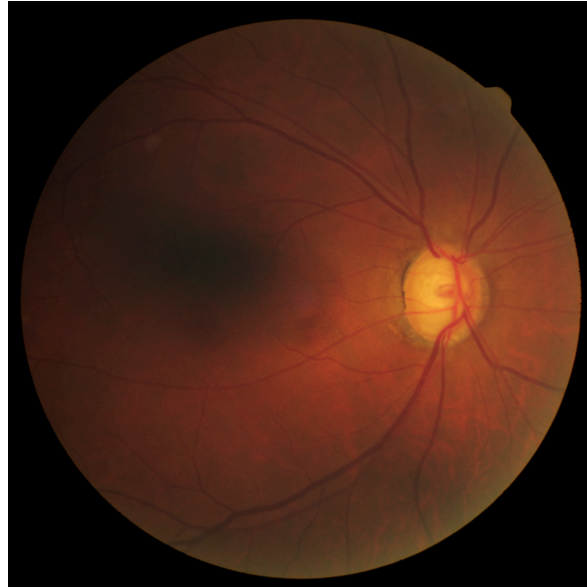


Figure 4. Retina with Glaucoma



Figure 5. Retina with Diabetic retinopathy

The shortage of eye specialists and required infrastructure in India causes retinal disease to go beyond the point where it can be treated easily or early. This often leaves treatment or surgery as the only option, but in rural areas, many people cannot afford these because of financial issues. This problem can be improved by regular and routine planning for eye care services in rural areas of India, such as organizing eye checkup camps. In these camps, we can have a few volunteers, a pilot (machine operator), and systems with Machine Learning (ML) and Deep Learning (DL) models that are capable of identifying retinal diseases on the spot. This makes the whole process cheaper, faster, and more efficient, and also helps people take necessary actions before their retinal conditions get worse or even lead to blindness.

1.1 Problem Definition

A major challenge in managing retinal diseases is detecting these conditions early, before they become serious and cause lasting damage. Many people in rural areas are not aware of the early warning signs for diseases like diabetic retinopathy, cataract, or glaucoma. Because of this lack of awareness, they often ignore mild symptoms and only visit a doctor once their vision problems have become severe. By then, it may be too late for simple treatments, and surgery might be the only option left.

If we can set up regular eye checkup camps in remote and rural areas, we could identify retinal diseases much earlier. Early detection would allow for more effective treatments, possibly preventing vision loss. We can also spread awareness through campaigns, teaching people about the early symptoms so they know when to seek medical help. This would help people take action before their condition worsens, ultimately saving both their vision and health.

Although machine learning (ML) and deep learning (DL) models seem to offer a solution for early detection, these methods require a large and balanced dataset with enough samples for each disease class. In practice, it is often difficult to collect such large and varied datasets, especially for medical images, because of privacy, cost, and logistics. Most available datasets have only a few classes and a limited number of images, making it challenging to train highly accurate models. Models trained on smaller datasets often fail to perform well in real-world, dynamic situations. This limitation restricts their ability to predict rare cases or subtle disease patterns. In our project, we have access to enough data for only four classes: Normal, Diabetic Retinopathy, Cataract, and Glaucoma.

1.2 Motivation

Accuracy in prediction and classification is extremely important when we use Machine Learning (ML) and Deep Learning (DL) in the medical field. If our models are not accurate, the chance of a wrong diagnosis increases, which can lead to poor decisions for patient care. When deciding between ML and DL, both have their uses, but deep learning models are usually better for medical image classification tasks. The main reason is that DL models, such as convolutional neural networks (CNNs), can automatically learn important features and patterns directly from the raw images, while traditional ML models usually need manually engineered features.

Deep learning is especially powerful in discovering complex and subtle patterns that may go unnoticed by ML algorithms. This advantage often leads to DL models outperforming traditional ML in the field of medical imaging. For this reason, DL is increasingly being chosen for tasks where feature extraction from images is critical, such as detecting retinal diseases.

In this report, we will explore both machine learning models—including K-Nearest Neighbors (KNN), Decision Tree, and Random Forest—and deep learning models like ResNet101, ResNet50, and DenseNet121. We will train and test these models on the same dataset so that we can fairly compare their performance. At the end, we will analyze their metrics—such as accuracy, F1-score,

and recall—to see which approach works best for retinal disease classification, with a special focus on why DL is generally more effective for medical image tasks.

1.3 Objective

The main goals of this project are:

1. To develop both ML and DL models and train them using the same dataset, allowing for a fair comparison between the two approaches.
2. To use different evaluation parameters—including accuracy, F1-score, recall, and others—to understand which type of model performs best.
3. To carefully analyze all results and provide a clear conclusion about which approach is better suited for retinal disease classification.

2. Literature Review

With advances in medical science and technology, the use of machine learning, deep learning, and artificial intelligence in healthcare is growing rapidly. These technologies are making health checkups faster, more efficient, and accessible to a larger number of people, especially in rural and remote areas. Early detection of diseases using computer-based methods allows doctors and healthcare workers to take necessary actions before conditions become serious, which can save lives and improve health outcomes.

The paper [1], "Machine Learning and Deep Learning-based Approach in Smart Healthcare: Recent Advances, Applications, Challenges, and Opportunities," (Rahman et. al., 2024) highlights the superior performance of deep learning—especially convolutional neural networks (CNNs)—in analyzing medical images. The authors show that deep learning models often outperform traditional machine learning methods for medical image classification.

The paper [2], "Machine Learning and Deep Learning Based Healthcare System: A Review," (Kumar A. and Kaur 2024) explains the basic ideas behind both ML and DL. It compares their methods and discusses how these technologies can be applied to different healthcare challenges. The authors point out that Deep Learning is particularly effective for medical image analysis because it can automatically extract important features from the data, reducing the need for hand-crafted features.

The paper [3], "A pilot cost-analysis study comparing AI-based EyeArt® and ophthalmologist assessment of diabetic retinopathy in minority women in Oslo, Norway" (Karabeg et. al., 2024), which performs a pilot cost-analysis study for detecting DR in a cohort of minority women with DM in Oslo, Norway, which has the highest prevalence of diabetes mellitus (DM) in the country, using both manual (ophthalmologist) and autonomous (AI) grading. This is the first study in Norway, as far as we know, that uses AI in DR-grading of retinal images.

The Paper [4], Review of machine learning techniques for glaucoma detection and prediction. The study (Khalil et. al. 2014) presented a survey to detect glaucoma changes through fundus images. Preprocessing, feature extraction, feature selection, and machine learning (ML) techniques were discussed.

The Paper [5], "Diagnosis of Diabetic Retinopathy using Machine Learning," Gupta and Karandikar (2015) reported a survey study that analyzes the automated techniques for DR diagnosis. A comparison was made between the algorithm that detects the various structural changes through fundus images.

The Paper [6], "Automated Detection and Diagnosis of Diabetic Retinopathy: A Comprehensive Survey", The survey (Lakshminarayanan et al. 2021) was conducted over five years, from 2016 to 2021, to investigate the automated techniques, which include ML and DL approaches, for the detection of DR in fundus and OCT images. A total of 114 papers were comprehensively reviewed from the open literature.

The Paper [7], A total of 114 papers were comprehensively reviewed from the open literature. Another review study (Abdul Lah et al. 2021) was presented that compared the automated ML techniques to detect the structural changes in fundus images. In addition to this, the author discussed the various fundus-related datasets (public and private).

The Paper [8], Automatic Detection of Diabetic Eye Disease Through Deep Learning Using Fundus Images: A Survey, Sarki et al. (2020), comprehensively reviewed state-of-the-art approaches for the detection of diabetic and glaucomatous changes through fundus images. Image processing, ML, and DL techniques were explored. The author also reported available datasets.

The Paper [9], The paper (Bala et al. 2021b) presented the clinical and technical survey for glaucoma diagnosis. It reported DL techniques for detecting pathological changes in fundus and OCT images.

3. MATERIALS AND METHODS

3.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a technique used by data scientists to explore and summarize data sets, primarily through visualization. It facilitates the discovery of patterns, identification of anomalies, hypothesis testing, and validation of assumptions, helping determine the optimal approach for data manipulation and analysis.

If we want to understand how common retinal diseases are in India, the research paper by (Gurudas et. al.,2024) analyzes the prevalence and risk factors of vision impairment and blindness among people with diabetes. Using data from the SMART-India study, which surveyed 42,147 adults aged 40 and older, the study provides national estimates. Participants with insufficient capacity to consent, pregnant women, and those with certain types of diabetes were excluded. Vision impairment and blindness were assessed using visual acuity standards, and diabetic retinopathy was graded using retinal images.

The analysis included 7,910 individuals with diabetes, 55.5% female and 44.5% male. The study found a national prevalence of 21.1% for vision impairment and 2.4% for blindness among diabetics. Those with ungradable images had higher impairment rates. Conditions like diabetic retinopathy significantly increase the risk of blindness. Risk factors identified included older age, low education, and unemployment.

The study estimates that, with 101 million diabetics in India as of 2021, around 21 million may have vision impairment, with 2.4 million being blind. It highlights the need for policymakers to target lower socio-economic groups to address healthcare disparities.

In the SMART-India study, out of 42,146 adults aged 40 and older, 7,910 individuals were identified with either known (5,689) or undiagnosed (2,221) diabetes. Of these, 55.5% were female. Retinal photographs were captured for all participants, with 22.5% being ungradable, 23.5% for known diabetes, and 19.7% for undiagnosed. Among the 6,133 participants with gradable images, 13.9% had some form of diabetic retinopathy, 6.0% had vision-threatening retinopathy, and 5.6% had diabetic macular edema.

Condition	Number of Participants	Percentage
Any Diabetic Retinopathy	855	13.9
Vision-Threatening Diabetic Retinopathy	365	6
Diabetic Macular Edema	345	5.6

Table 1. Retinopathy Prevalence Among Diabetics

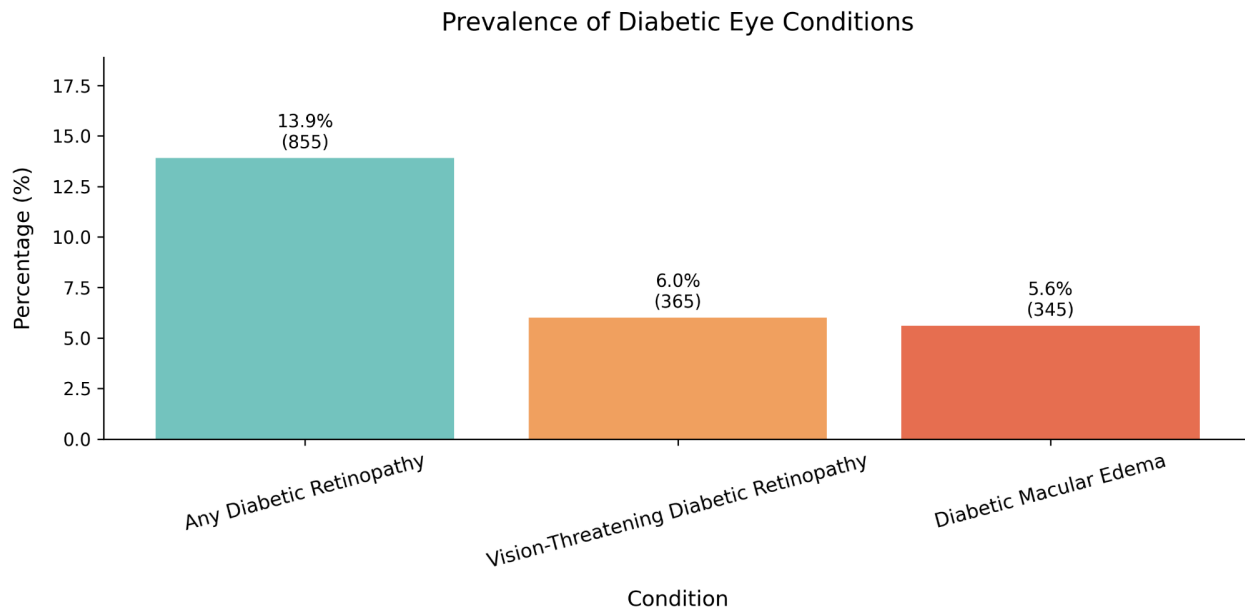


Figure 6: Retinopathy Prevalence Among Diabetics

This is only about retinal diseases related to diabetes, where the numbers are high because of diabetic cases. If we also look at cataract, glaucoma, and other issues in non-diabetic patients, the numbers will be even higher. This shows the risk of how many Indians are prone to retinal diseases, and since the numbers are rising, it's important to get a proper diagnosis so it can be prevented before it becomes critical.

If we want to understand how common retinal diseases are in India, the research paper by (Kovai et.al., 2007) provides valuable insights. In their study, 5,573 people above the age of 15 from Andhra Pradesh were interviewed and examined. Out of these, 1,234 individuals (22.1%) had distant visual acuity less than 20/60 or equivalent field loss in the better eye. Among those affected, 898 (72.7%) had not sought treatment despite noticing decreased vision, mainly due to personal, economic, and social reasons. The study found that literate individuals, those classified as blind, and those with cataract or other causes of visual impairment were more likely to seek treatment. Common barriers for not seeking treatment included personal reasons (52%), economic constraints (37%), and social reasons (21%). Routine planning for eye care services in rural areas of India must address the barriers to eye care perceived by communities to increase the utilization of services.

Category	Count	Percent	Notes
Visual Acuity < 20/60 or Field Loss	1234	22.1	Of the total interviewed
Did Not Seek Treatment	898	72.7	Of those with VA < 20/60

Table 2. Visual Impairment and Treatment Gaps, (Total Interviewed Out of 5573).

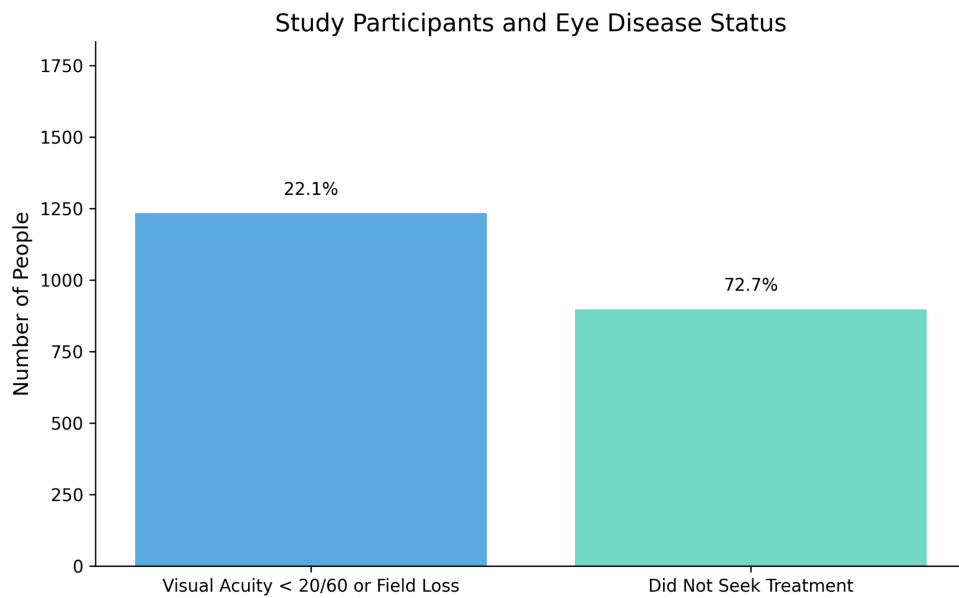


Figure 7. Visual Impairment and Treatment Gaps, (Total Interviewed Out of 5573).

Barrier Type	Percent
Personal Barriers	52
Economic Barriers	37
Social Barriers	21

Table 3. Causes of People who are not seeking treatment

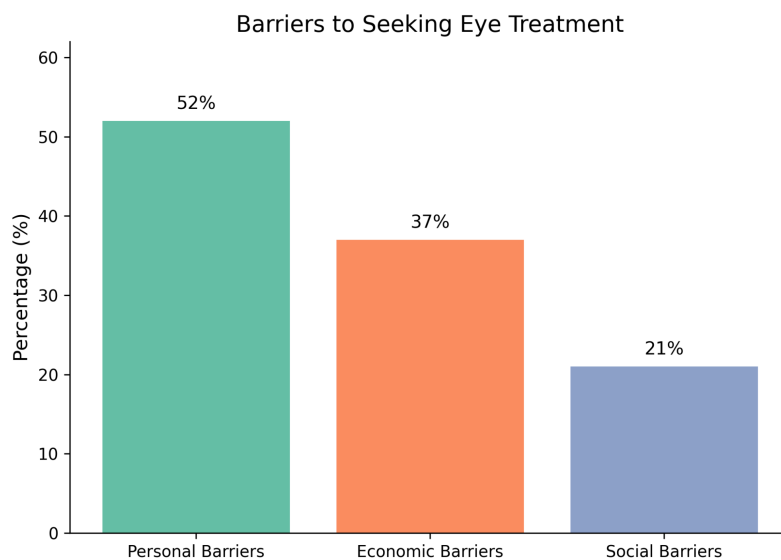


Figure 8. Causes of People who are not seeking treatment

3.2 Dataset Details

The initial phase of this AI-based retinal disease diagnosis project focuses on robust data collection from authentic and verified sources. For this purpose, we utilized a comprehensive dataset available on Kaggle, which is particularly suited to our requirements. This dataset contains retinal images categorized into four primary classes: Normal, Diabetic Retinopathy, Cataract, and Glaucoma, with each class comprising approximately 1000 images. These images are curated from various reputable sources, such as IDRiD, Oculus Recognition, and HRF. During the dataset selection process, we found that while multiple retinal image datasets exist on Kaggle and other repositories, most suffer from class imbalance—either lacking sufficient images per class or providing uneven class distribution. Such issues can severely hinder the training of machine learning (ML) and deep learning (DL) models, leading to poor generalization, inaccurate classification, and biases in prediction that are often revealed through analysis such as confusion matrices. (<https://www.kaggle.com/datasets/gunavenkatdoddi/eye-diseases-classification>).



Figure 9. Different stages are involved in constructing a learning model.

The selected dataset addresses these limitations by ensuring nearly equal representation for each disease class. It is organized into pre-labeled folders, with each folder dedicated to a specific class (retinal disease), streamlining the data preprocessing and loading process for model training. This structure not only simplifies implementation but also helps prevent model bias towards any particular class, thereby promoting reliable performance during both training and evaluation phases. The dataset comprises color fundus retinal images—high-resolution photographs of the interior surface of the eye, including the retina, optic disc, macula, and retinal blood vessels. These images are captured using a specialized low-power microscope with an attached camera, known as a fundus camera. Color fundus images are widely used in ophthalmology because they vividly illustrate pathological changes associated with various retinal diseases. The color information captured in these images is critical for diagnosing conditions like diabetic retinopathy, cataract, and glaucoma, as it reveals details about hemorrhages, exudates, vessel abnormalities, and other key biomarkers. By leveraging such color fundus images with balanced class representation, our dataset provides a strong foundation for developing deep learning models capable of effective and accurate multi-class retinal disease detection, tailored for deployment in rural screening camps and low-resource settings.

SL. NO.	Name of the Retinal disease	No. of Images	Percentage (%)
1	(N) Normal Fundus	1074	24.61
2	(DR) Diabetic Retinopathy	1098	26.04
3	(C) Cataract	1038	24.61
4	(G) Glaucoma	1007	23.88

Table 4. Retinal disease in the dataset, number of images.

Although image quality is not a major issue, since we typically resize images before training to reduce their size and speed up the training process, the quantity of images is crucial. In previous attempts, I used images

from multiple classes with significantly different numbers of samples; sometimes one class had only hundreds of images while another had thousands. This imbalance negatively affects the training of ML and DL models, as the model tends to be dominated by the class with more images. As a result, the model may "hallucinate" or perform poorly, struggling to differentiate between classes because it has not been trained on an equal number of samples per class, or some classes have larger samples of data, but some have smaller samples of data. This poor performance becomes evident when analyzing the model's results using a confusion matrix, which we can observe below. The previous dataset was also from Kaggle. <https://www.kaggle.com/datasets/ruhulaminsharif/eye-disease-image-dataset>.

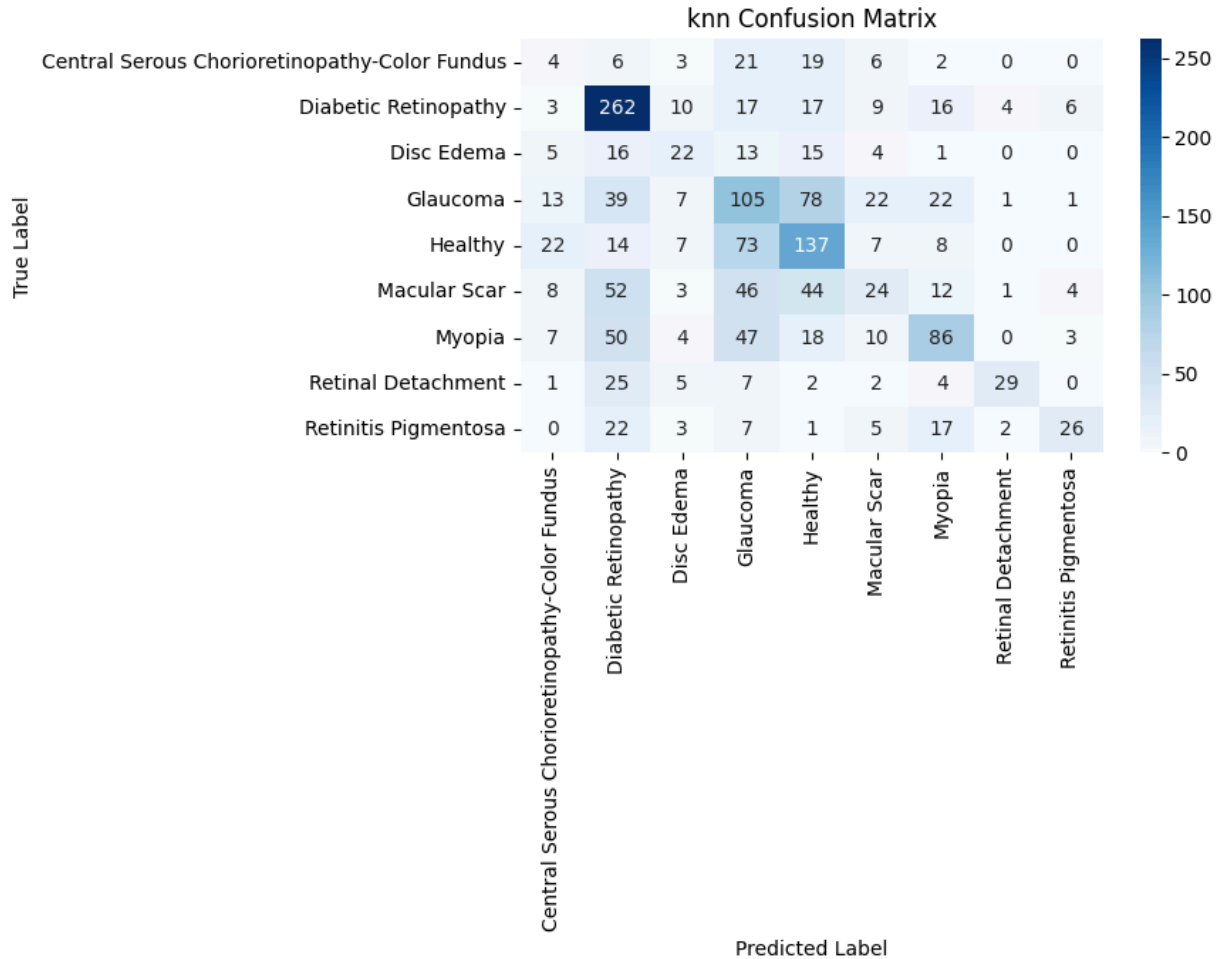


Figure 10: KNN Confusion Matrix from Previous Dataset

3.3 CLASSIFICATION ALGORITHMS

In this project, our goal is to build a system that can automatically identify and classify different retinal diseases from fundus images. To do this, we apply a variety of classification techniques, which fall mainly under two major categories: Machine Learning (ML) algorithms and Deep Learning (DL) models. Each of these techniques has its strengths and a unique way of working with data. By testing and analyzing both, we can get a clear idea about which approach is most effective for medical image classification, specifically in the context of retinal disease detection.

For our classification analysis, we specifically selected three popular and widely-used Machine Learning algorithms:

1. K-Nearest Neighbors (KNN)
2. Decision Tree
3. Random Forest

These methods were chosen because they are commonly used for classification tasks and have proven their effectiveness in various domains, including the medical field. Machine Learning algorithms generally require feature extraction, where meaningful characteristics or measurements are derived from the images, which then act as inputs for the model. ML models are usually less demanding in terms of computational requirements and can work well with smaller datasets.

On the other hand, Deep Learning models have recently gained a lot of attention for image classification tasks because they can automatically learn features directly from raw images without the need for manual intervention. Deep Learning, especially Convolutional Neural Networks (CNNs), has shown great success in various medical imaging applications, outperforming traditional ML models in many cases.

For the Deep Learning side of our analysis, we selected three well-known CNN architectures:

1. ResNet101
2. ResNet50
3. DenseNet121

These models have been carefully chosen due to their strong performance on image datasets and their ability to learn complex features. They are also widely used in the medical imaging community for tasks such as disease detection, classification, and segmentation.

By using both ML and DL models, we aim to cover the strengths of each approach. The Machine Learning models help us understand the effectiveness of simple, interpretable algorithms, while the Deep Learning models illustrate the power of automatically extracting features and capturing intricate patterns from the images. This two-pronged approach enables a thorough comparison of not only their accuracy, but also other key metrics such as F1-score, recall, precision, specificity, and efficiency.

3.3.1 K-Nearest Neighbor (KNN)

K-Nearest Neighbors (KNN) (Pathak et. al, 2024) is a simple and popular machine learning algorithm for classification tasks. When KNN is given a new data point to classify, it looks at the ‘K’ closest data points from its training set. These are called its “neighbors.” After finding these neighbors, the algorithm counts which class appears most often among them and assigns that class to the new data point.

KNN does not build a model during training. Instead, it memorizes the whole training dataset and waits until it needs to make a prediction. Then, it searches through all the data to find the nearest neighbors. Because of

this, KNN is easy to understand and implement, but it can become slow and less accurate if there are too many data points or if the data has many features (dimensions). It works best with smaller, balanced datasets.

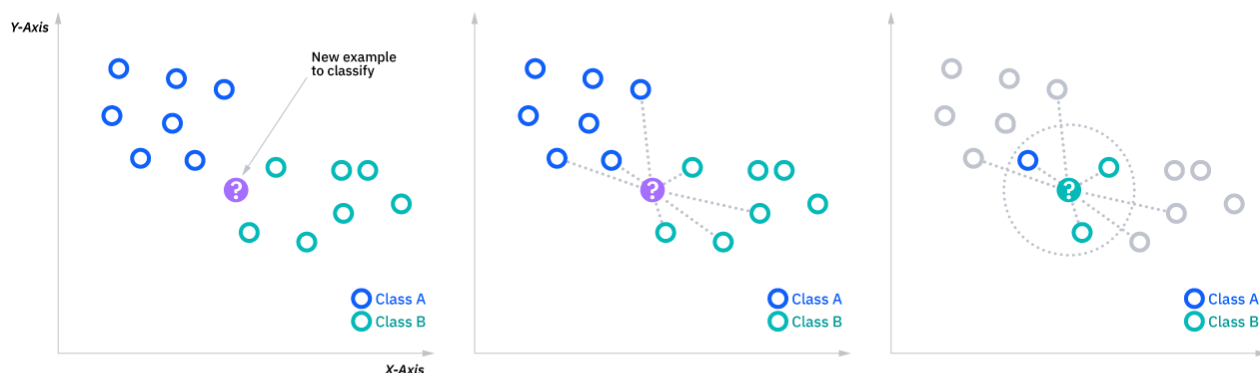


Figure 11: Understanding KNN with Diagram

3.3.2 Decision Tree

A flowchart-like structure that is used to make decisions or predictions based on data. It works by splitting the data into smaller groups based on a series of decision rules. At each node in the tree, a rule is applied that splits the data further. This process continues until the tree reaches the final prediction at the leaf node. While decision trees are easy to understand and quick to run, they often struggle with complex data because they cannot automatically learn and extract deep features like modern deep learning algorithms can.

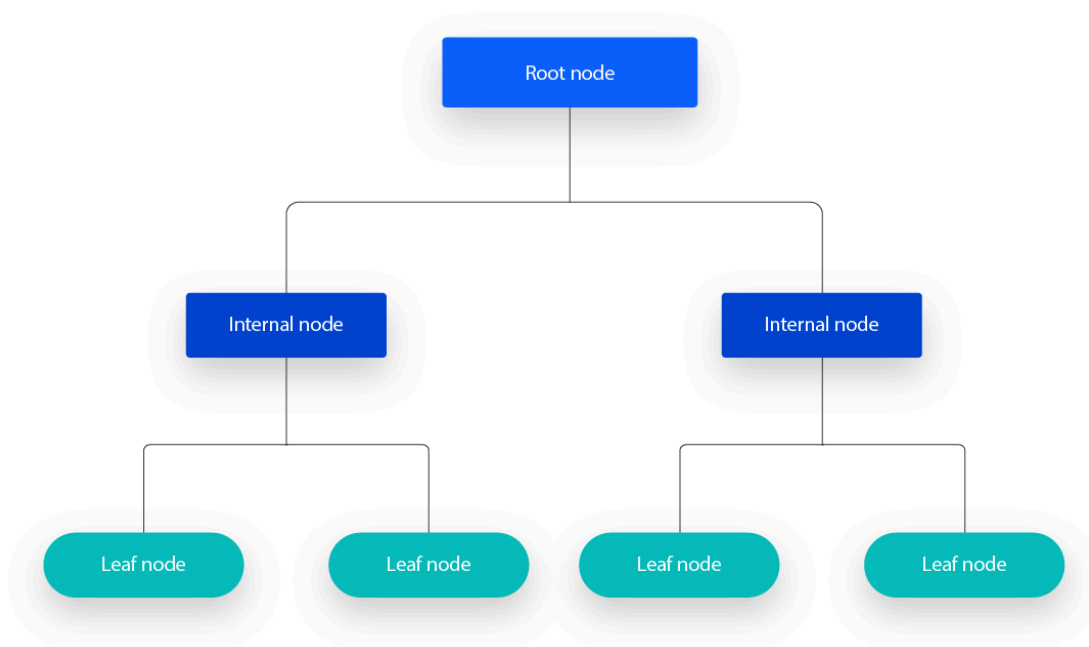


Figure 12: Understanding Decision Tree with Diagram

3.3.3 Random forest

It's a popular and powerful machine learning algorithm used for classification and regression tasks. It works by combining the predictions of many decision trees (forming a "forest") to make a final decision. In a

random forest, each decision tree is trained on a random subset of the data and uses a random subset of the features when splitting nodes. When it is time to make predictions, each tree gives a result, and the one that gets the most votes becomes the final output (for classification).

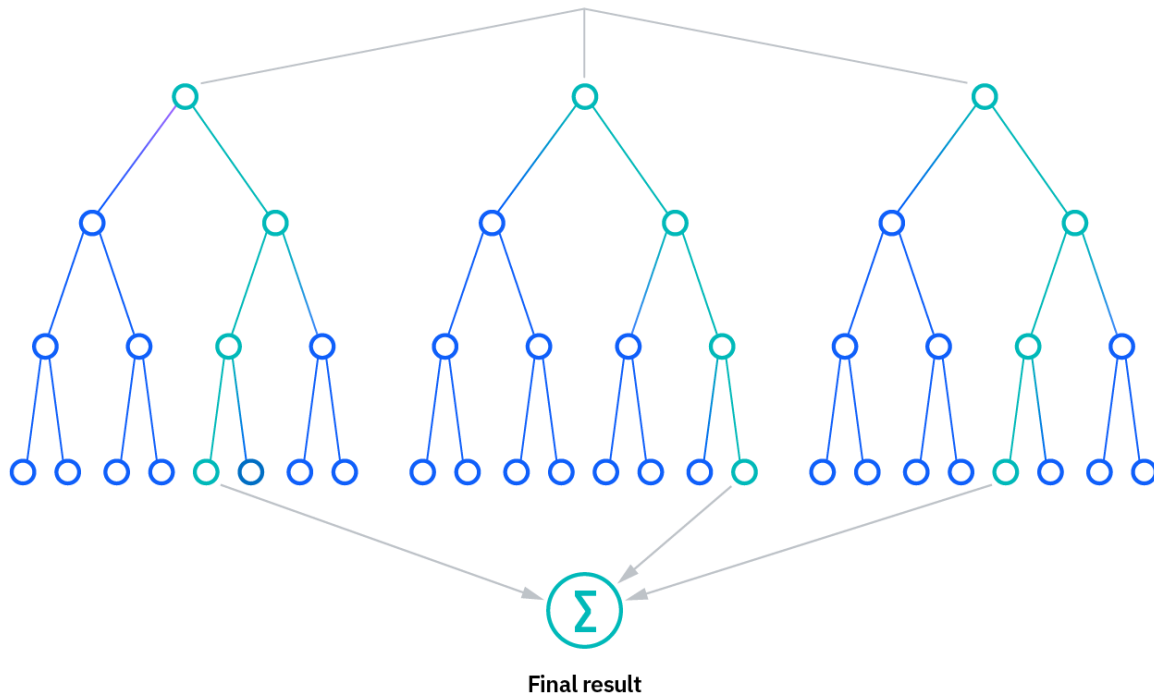


Figure 13: Understanding Random Forest

Random Forests usually provide better accuracy than a single decision tree because they reduce the risk of overfitting and can handle complex data better. They are widely used in many tasks, including image classification, due to their robustness and flexibility.

3.3.4 CNN

A Convolutional Neural Network (CNN) is a type of deep learning model specifically designed for processing and analyzing image data. CNNs are widely used in image classification, object detection, and many other computer vision tasks. The key component of a CNN is the convolutional layer. This layer uses small filters (also called kernels) that slide over the input image, looking for specific patterns like edges, textures, or shapes. The purpose of the convolutional layer is to automatically extract important features from images, which helps the network understand and recognize visual patterns. By stacking multiple convolutional layers, CNNs can learn increasingly complex features, making them highly effective for image-related tasks.

3.3.5 How does CNN work?

A Convolutional Neural Network (CNN) is especially effective for analyzing images, audio, and speech data. It is built from three main types of layers: convolutional layers, pooling layers, and fully-connected (FC) layers. The convolutional layer, which comes first, detects simple features like edges or colors. As the data passes through additional layers—alternating between convolutional and pooling layers—the network gradually learns to recognize more complex patterns and larger structures within the image. Finally, the fully-connected layer (at the end) uses all these learned features to identify or classify the object in the image. This layered processing enables CNNs to achieve high performance in visual recognition tasks.

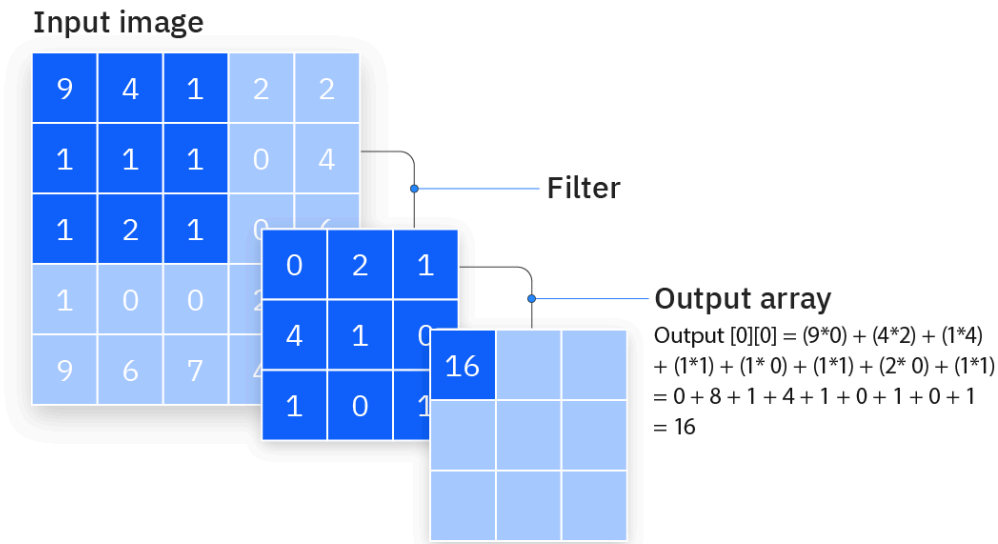


Figure 14: Understanding CNN with Diagram

3.3.6 ResNet101

ResNet101 is a deep convolutional neural network (CNN) architecture that is widely used for image classification and related tasks. It is part of the Residual Network (ResNet) family, which introduced the concept of residual connections to help train very deep neural networks efficiently. The "101" in ResNet101 indicates that the network has 101 layers. ResNet101 uses shortcut (residual) connections that skip one or more layers, allowing the network to learn the difference (the "residual") between layers instead of trying to learn the full transformation. This approach helps solve the problems of vanishing and exploding gradients, making it possible to train much deeper and more accurate networks.

3.3.7 ResNet50

ResNet50 is a deep convolutional neural network that is widely used for image classification and recognition tasks. It is a member of the ResNet (Residual Network) family, which introduced the concept of "residual connections" or "shortcut connections" to address the challenge of training very deep neural networks. The "50" in ResNet50 refers to the number of layers in the network. ResNet50 uses shortcut connections to allow the network to learn residual functions, which help improve accuracy by making it easier to train deeper models. This architecture enables ResNet50 to extract and learn complex features from images efficiently.

3.3.8 DenseNet121

DenseNet121 is a deep convolutional neural network that is widely used for image classification and computer vision tasks. The name "DenseNet" comes from its unique architecture, where each layer is directly connected to every other layer in a dense fashion. This means that the output of each layer is passed as input to all subsequent layers within the same block.

The “121” in DenseNet121 refers to the total number of layers in the network. This dense connectivity helps improve the flow of information and gradients through the network, making it easier to train and leading to better performance with fewer parameters compared to traditional architectures.

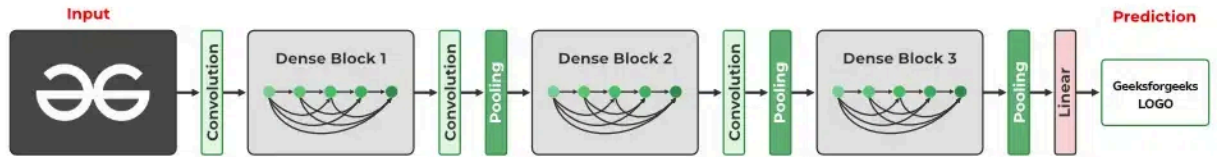


Figure 15: Understanding DenseNet121 with Diagram.

3.4 Performance Metrics

Some suggested and commonly used metrics for evaluating machine learning and deep learning models, especially for classification tasks such as image classification. These metrics help you objectively assess the performance of your classification model from various perspectives, ensuring reliable and interpretable results.

3.4.1 Accuracy

Accuracy is a basic yet important evaluation metric used to measure the performance of classification models. In the context of medical image classification, such as diagnosing retinal diseases by analyzing medical images, accuracy tells you how often the model correctly classifies images compared to the total number of classifications it makes.

$$\text{Accuracy} = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

Formula Figure 1. Formula of Accuracy

3.4.2 Precision

Precision measures the accuracy of positive predictions. Specifically, it tells us what proportion of items labeled as positive by the model are positive according to the ground truth.

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

Formula Figure 2. Formula of Precision

3.4.3 Recall

Recall (also called sensitivity or true positive rate) measures the ability of a model to find all relevant (positive) cases in your data

$$\text{Recall (or TPR)} = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

Formula Figure 3. Formula of Recall

3.4.4 Specificity

Specificity (also called the true negative rate) measures a model's ability to correctly identify negatives—that is, how well the model avoids false alarms.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Formula Figure 4. Formula of Specificity

3.4.5 F1 Score

The F1 score is the harmonic mean of precision and recall. It captures both the precision (how many of the positive predictions were correct) and recall (how many actual positives the model found), providing a single number that balances the two.

$$F_1 = \left(\frac{\text{recall}^{-1} + \text{precision}^{-1}}{2} \right)^{-1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Formula Figure 5. Formula of F1 Score

3.4.6 FPR

FPR measures the proportion of actual negatives that are incorrectly classified as positives. In medical terms, the rate at which healthy patients are wrongly diagnosed as diseased.

$$\text{FPR} = \frac{\text{incorrectly classified actual negatives}}{\text{all actual negatives}} = \frac{FP}{FP + TN}$$

Formula Figure 5. Formula of FPR

3.4.7 FNR

FNR measures the proportion of actual positives that are incorrectly classified as negatives. In medical terms, the rate at which diseased patients are missed (not diagnosed) by the model.

$$\text{FNR} = \frac{FN}{FN + TP}$$

Formula Figure 6. Formula of FNR

4. EXPERIMENTAL RESULTS

4.1 KNN

Class	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
cataract	91.23	77.69	90.38	91.51	83.56	8.49	9.62
diabetic_retinopathy	81.99	64.41	69.09	86.54	66.67	13.46	30.91
glaucoma	85.31	74.07	59.41	93.46	65.93	6.54	40.59
normal	81.28	63.73	60.75	88.25	62.2	11.75	39.25
Macro Average	84.95	69.97	69.91	89.94	69.59	10.06	30.09

Table 5. KNN Algorithm Overall Performance.

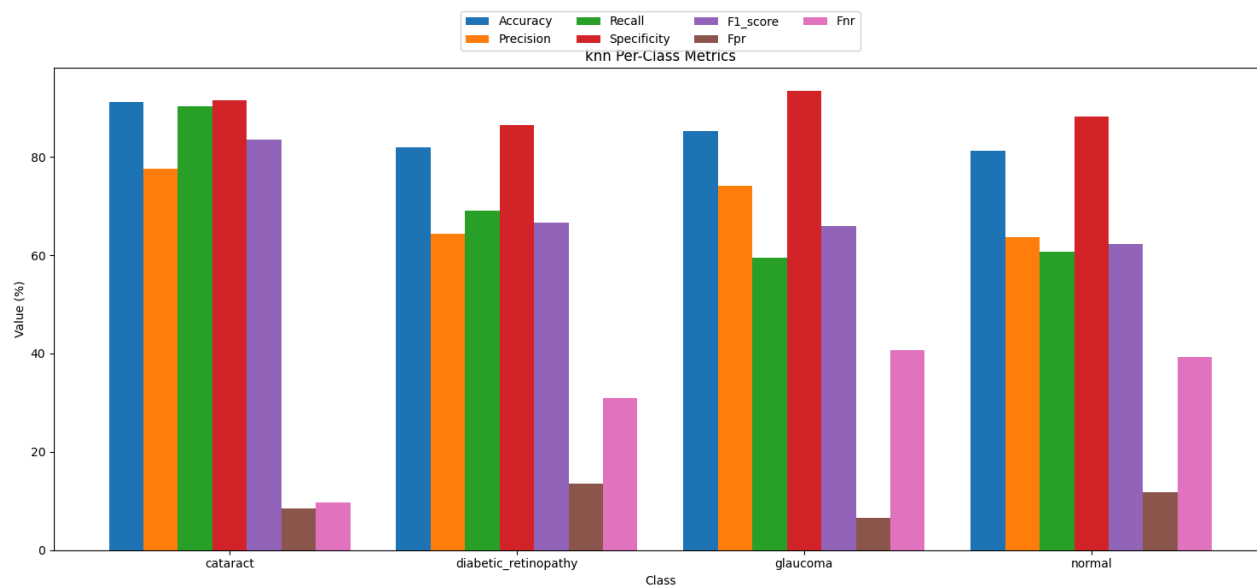
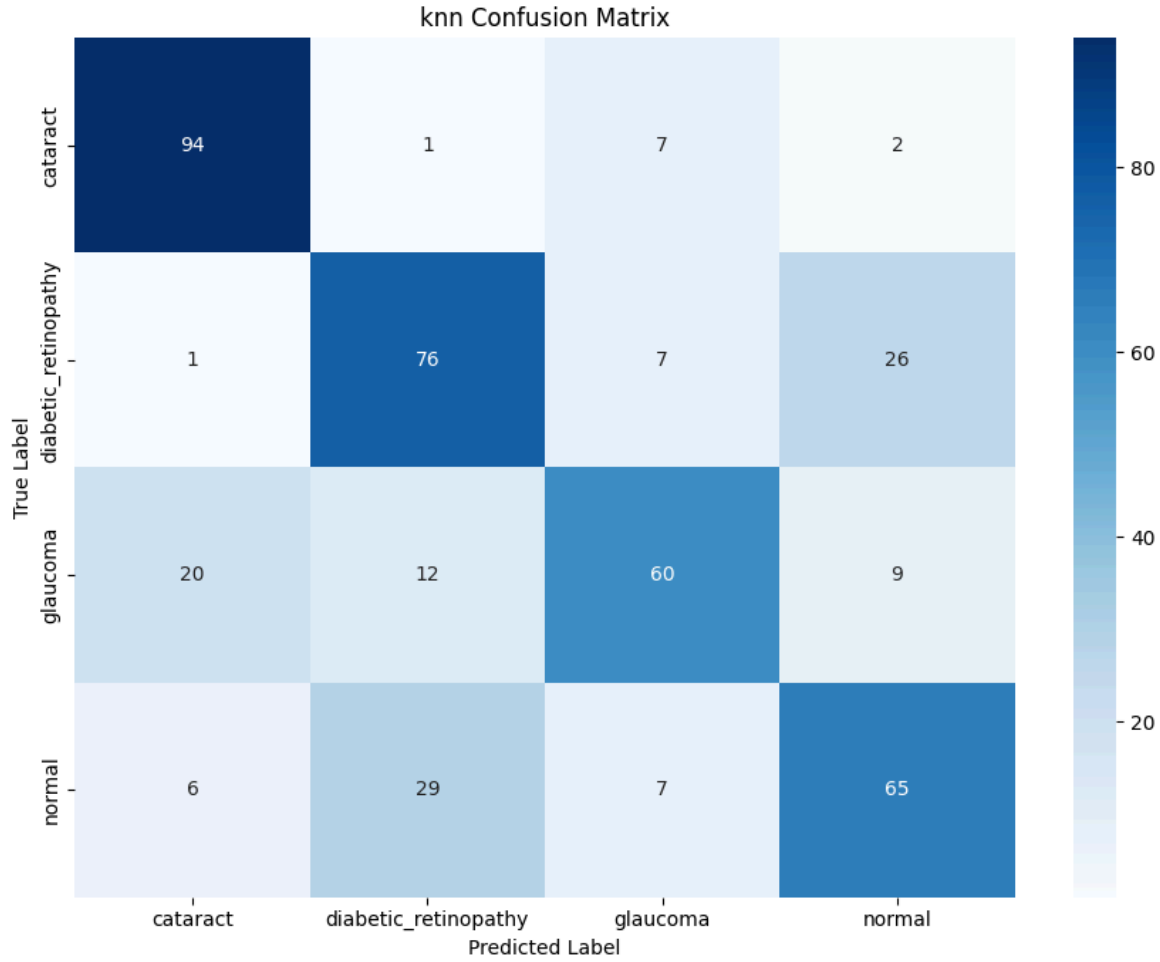


Figure 16. KNN Algorithm Overall Performance.

**Figure 17. KNN Confusion Matrix**

4.2 Decision Tree

Class	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
cataract	84.36	68.27	68.27	89.62	68.27	10.38	31.73
diabetic_retinopathy	77.25	56.25	57.27	84.29	56.76	15.71	42.73
glaucoma	75.59	49.04	50.5	83.49	49.76	16.51	49.5
normal	72.27	45.1	42.99	82.22	44.02	17.78	57.01
Macro Average	77.37	54.66	54.76	84.91	54.7	15.09	45.24

Table 6. Decision Tree Algorithm Overall Performance

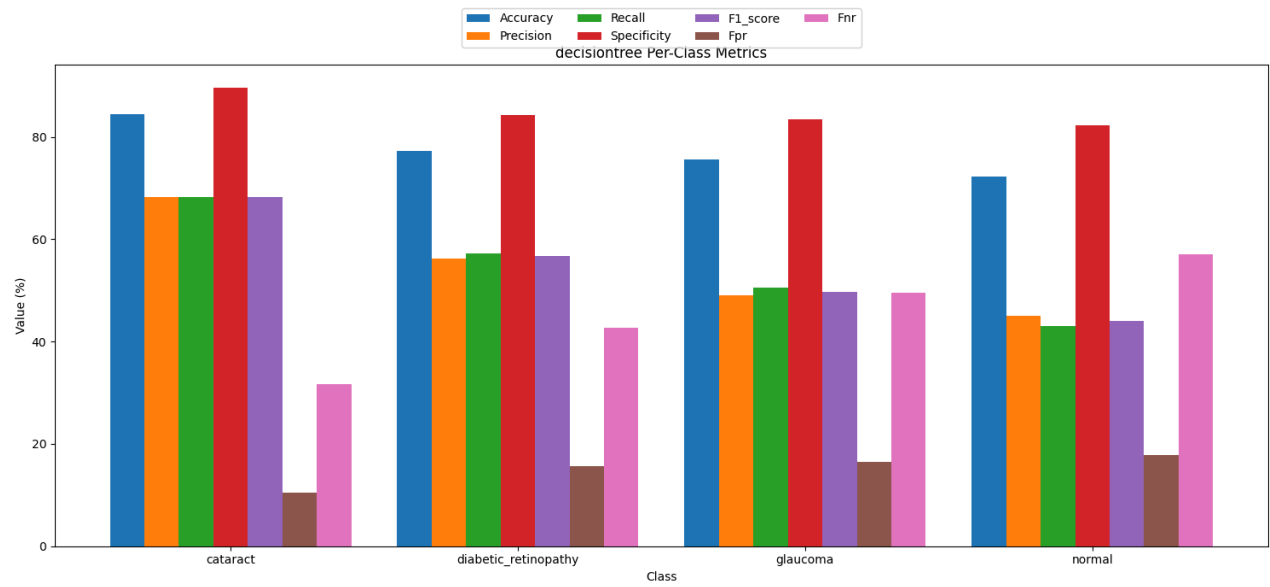
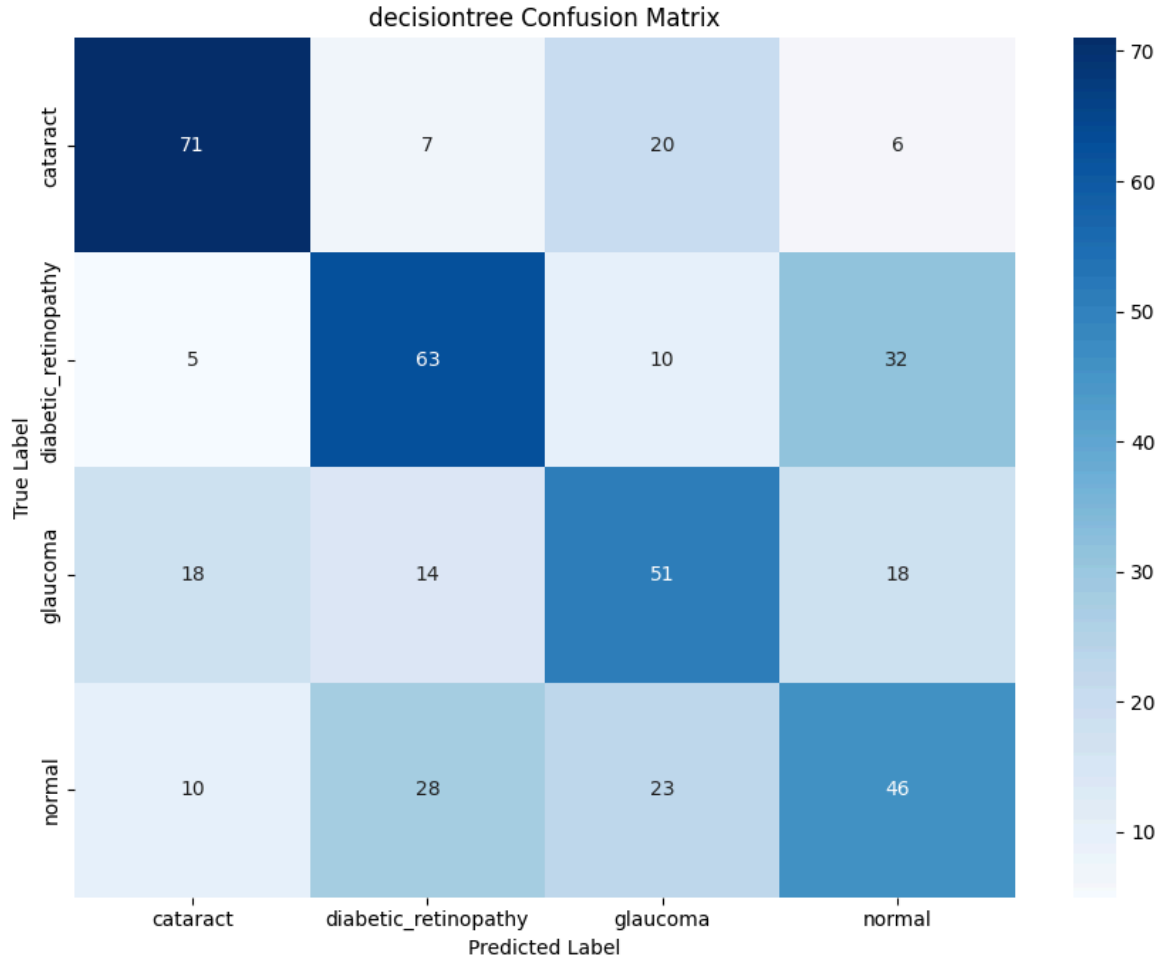


Figure 18. Decision Tree Algorithm Overall Performance

**Figure 19. Decision Tree Confusion Matrix**

4.3 Random Forest

Class	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
cataract	90.52	78.57	84.62	92.45	81.48	7.55	15.38
diabetic_retinopathy	86.02	75.76	68.18	92.31	71.77	7.69	31.82
glaucoma	81.04	60	62.38	86.92	61.17	13.08	37.62
normal	79.38	59.43	58.88	86.35	59.15	13.65	41.12
Macro Average	84.24	68.44	68.51	89.51	68.39	10.49	31.49

Table 7. Random Forest Algorithm Overall Performance

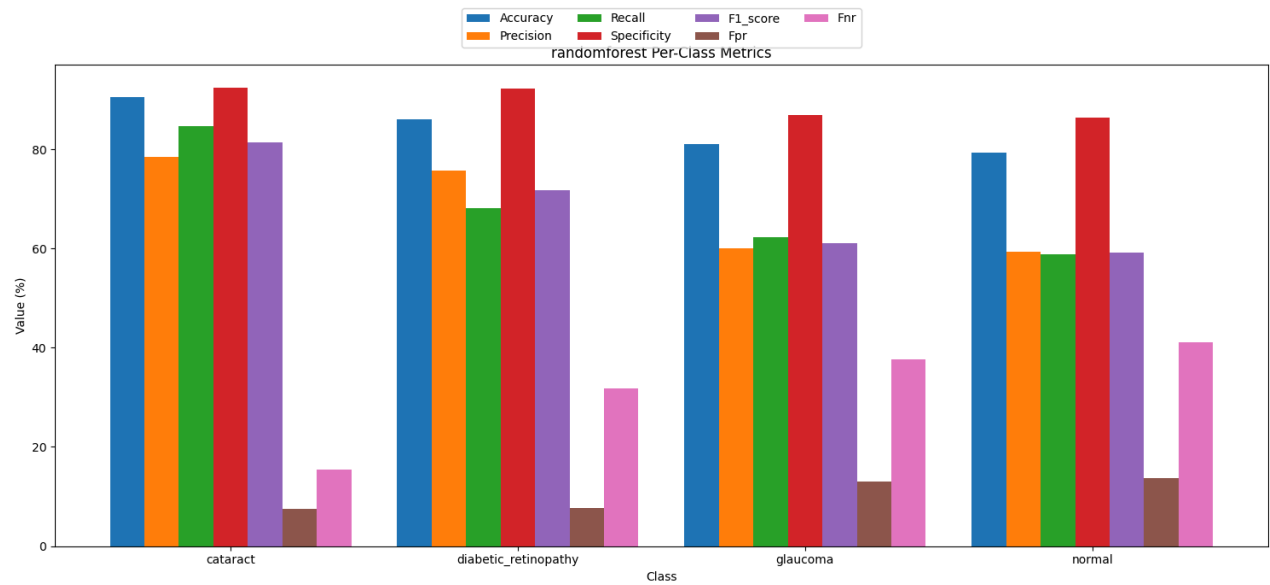


Figure 20. Random Forest Algorithm Overall Performance

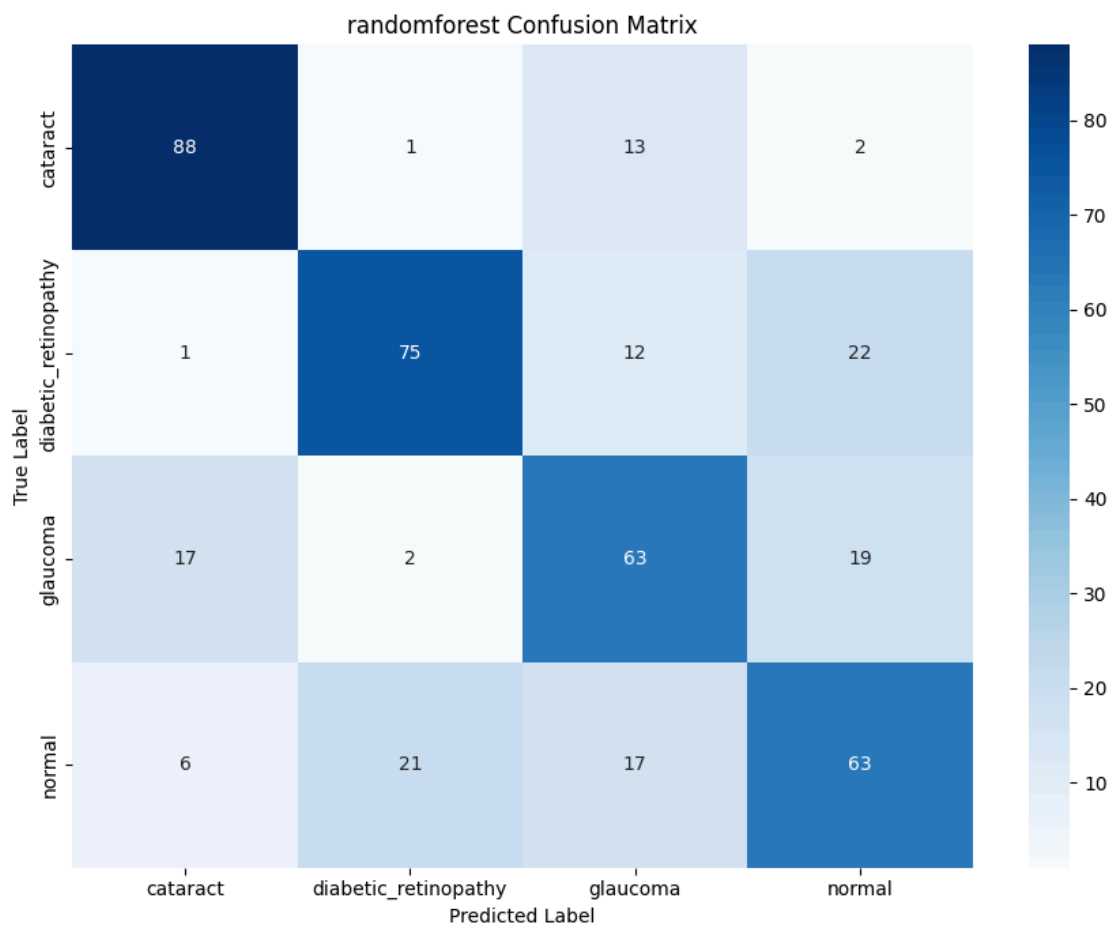


Figure 21. Random Forest Confusion Matrix

4.4 ResNet101

Class	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
cataract	95.27	87.3	96.49	94.82	91.67	5.18	3.51
diabetic_retinopathy	100	100	100	100	100	0	0
glaucoma	93.62	87.34	80.23	97.03	83.64	2.97	19.77
normal	95.51	93.46	89.29	97.75	91.32	2.25	10.71
Macro Average	96.1	92.03	91.5	97.4	91.66	2.6	8.5

Table 8. ResNet101 Algorithm Overall Performance

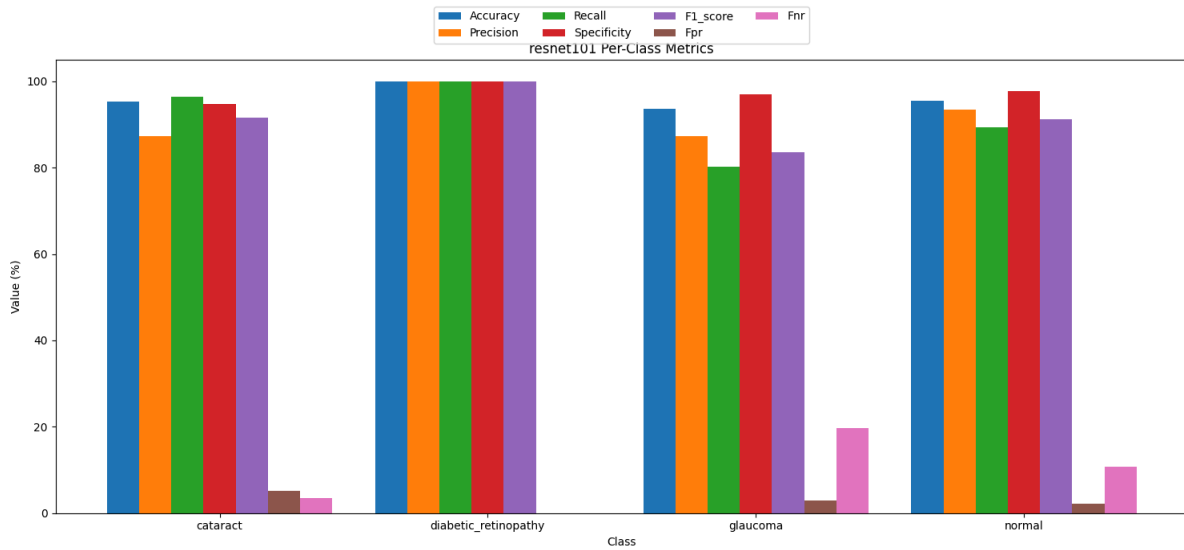


Figure 22. ResNet101 Algorithm Overall Performance

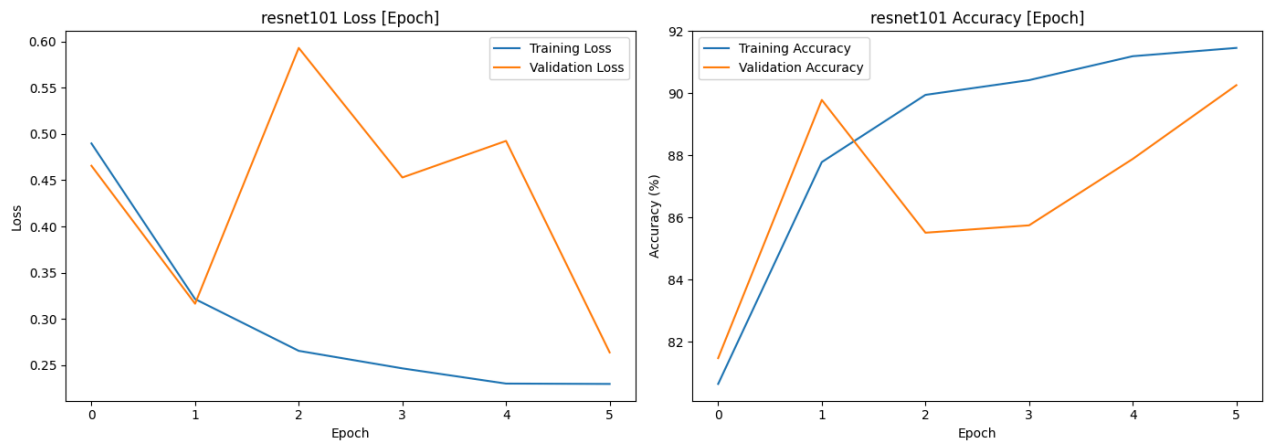
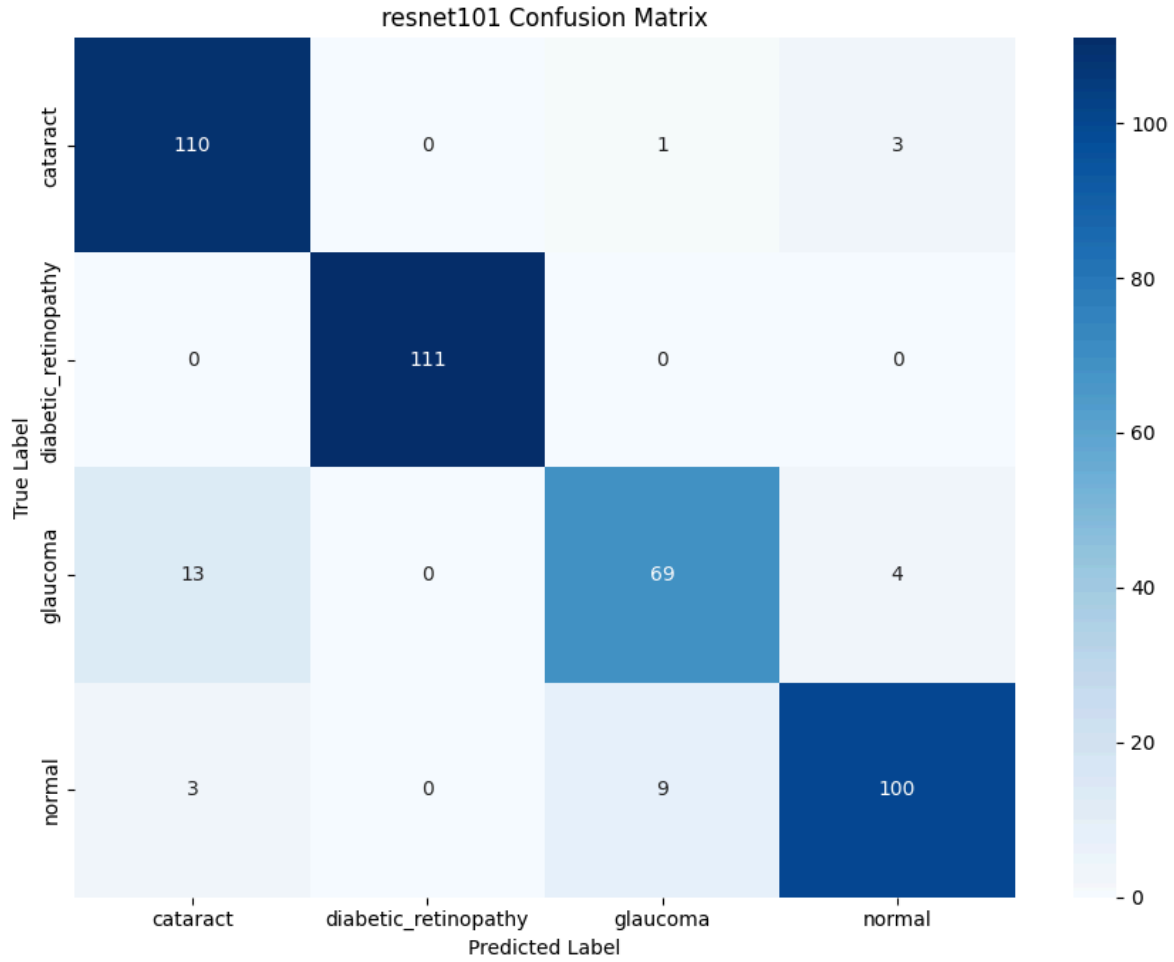


Figure 23. ResNet101 Algorithm, Loss, and Accuracy over Epoch.

**Figure 24. ResNet101 Confusion Matrix**

4.5 ResNet50

Class	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
cataract	95.74	95.28	88.6	98.38	91.82	1.62	11.4
diabetic_retinopathy	99.76	99.11	100	99.68	99.55	0.32	0
glaucoma	94.33	84.44	88.37	95.85	86.36	4.15	11.63
normal	95.04	89.57	91.96	96.14	90.75	3.86	8.04
Macro Average	96.22	92.1	92.23	97.51	92.12	2.49	7.77

Table 9. ResNet50 Algorithm Overall Performance

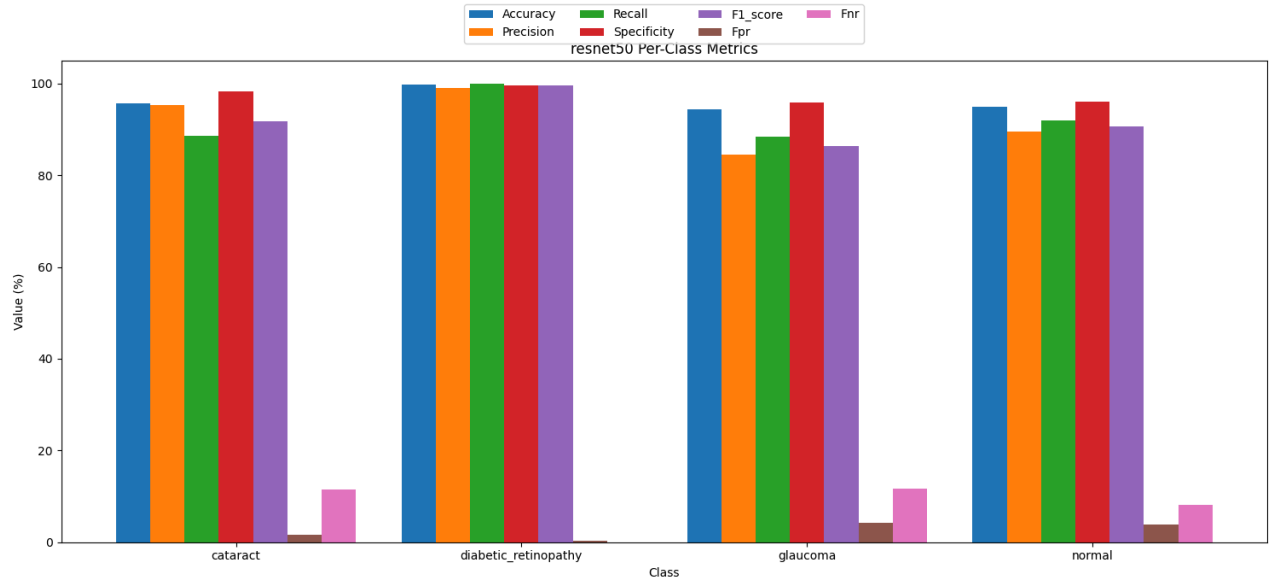
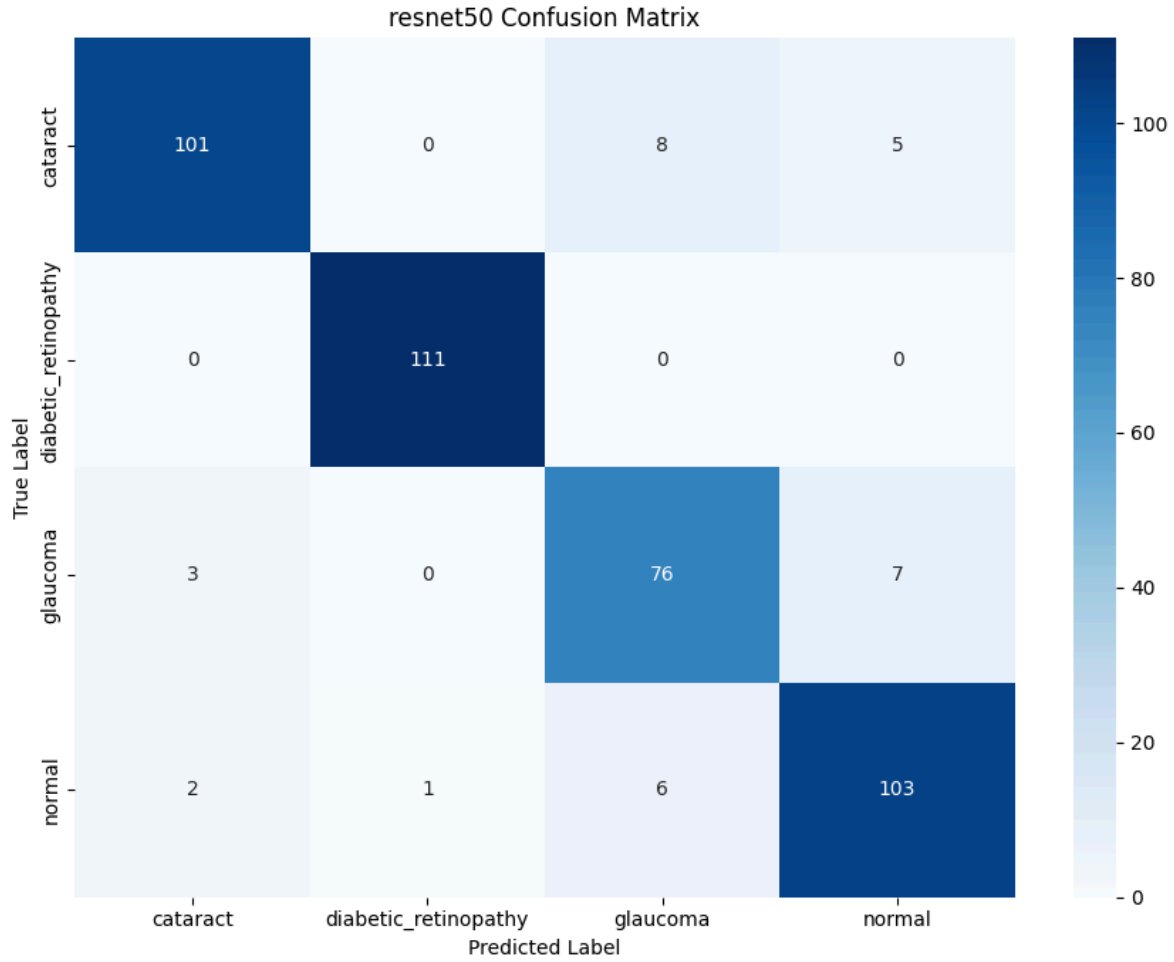


Figure 25: ResNet50 Algorithm Overall Performance



Figure 26. ResNet50 Algorithm, Loss, and Accuracy over Epoch.

**Figure 27. ResNet50 Confusion Matrix**

4.6 DenseNet121

Class	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
cataract	92.43	81.06	93.86	91.91	86.99	8.09	6.14
diabetic_retinopathy	100	100	100	100	100	0	0
glaucoma	91.96	87.14	70.93	97.33	78.21	2.67	29.07
normal	94.33	90	88.39	96.46	89.19	3.54	11.61
Macro Average	94.68	89.55	88.3	96.43	88.6	3.57	11.7

Table 10. DenseNet121 Algorithm Overall Performance

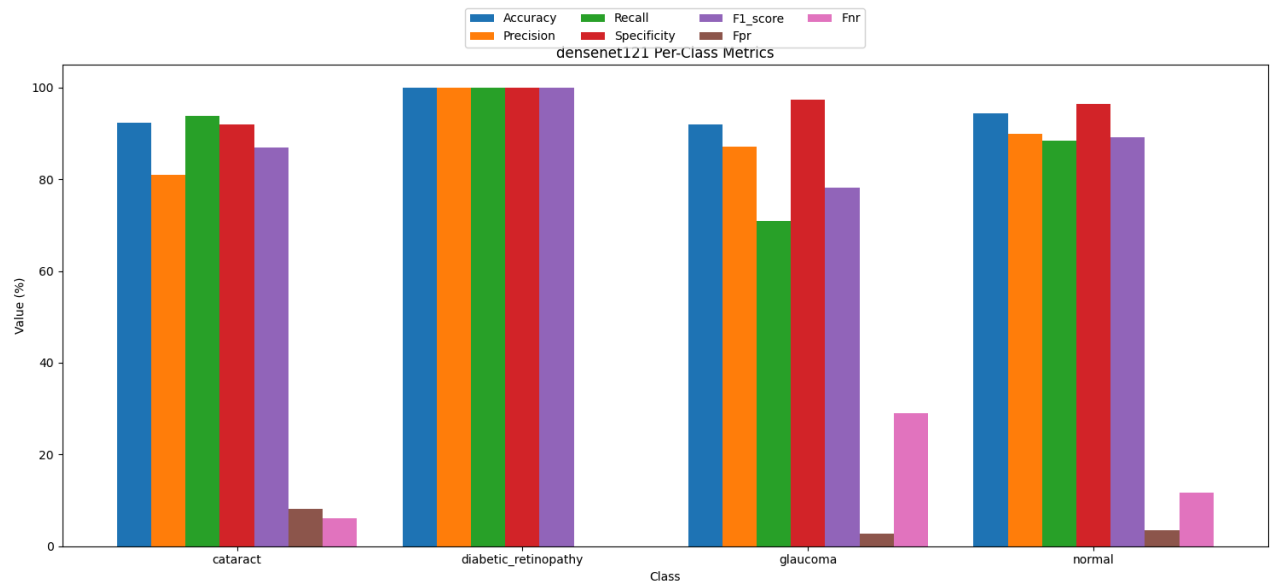


Figure 28. DenseNet121 Algorithm Overall Performance

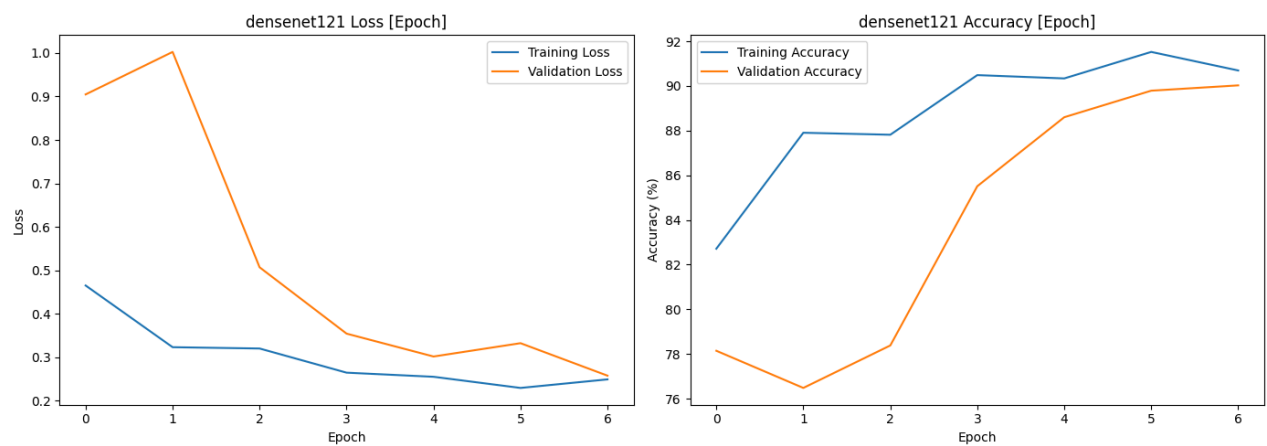


Figure 29. DenseNet121 Algorithm, Loss, and Accuracy over Epoch.

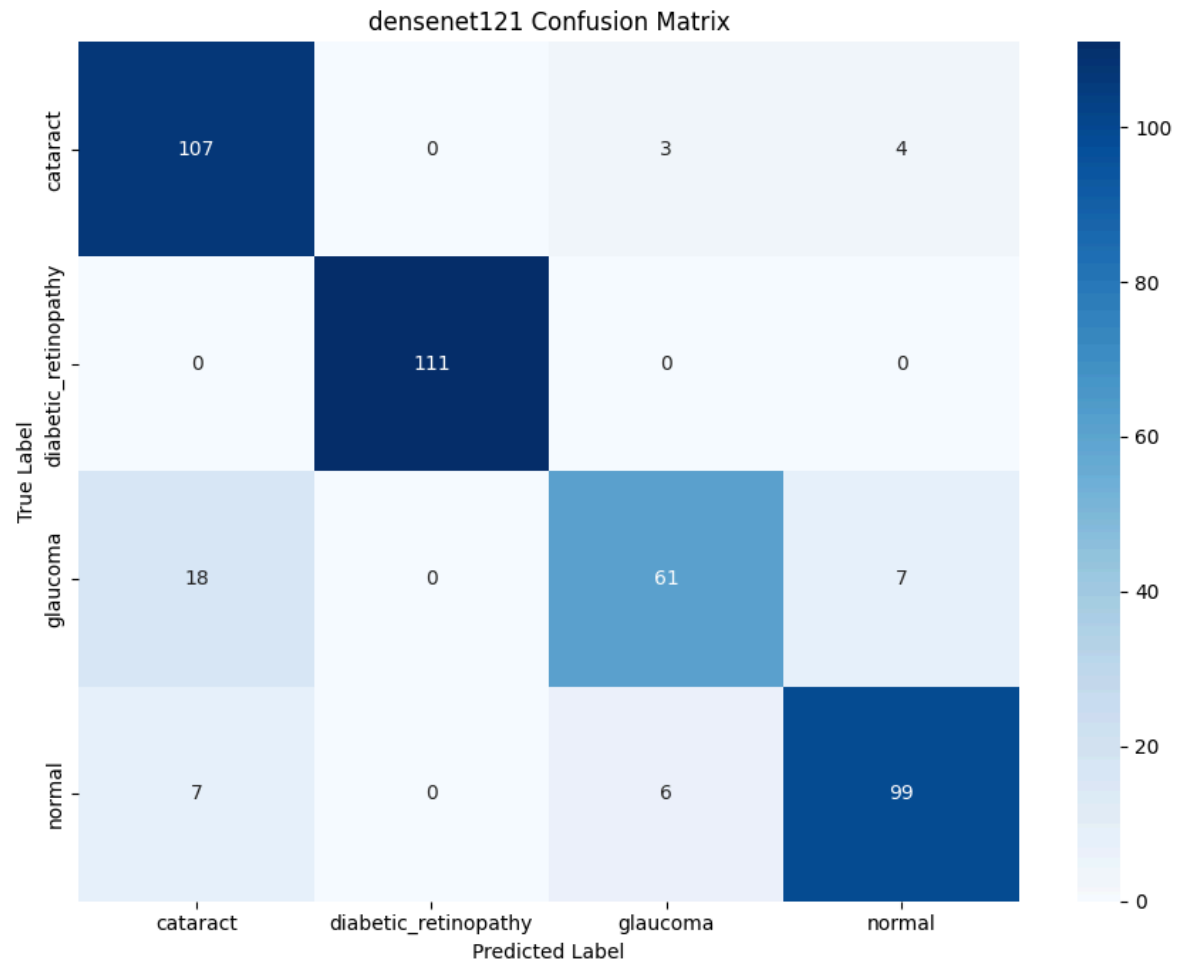


Figure 30. DenseNet121 Confusion Matrix

Model	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
Decision Tree	77.37	54.66	54.76	84.91	54.7	15.09	45.24
Random Forest	84.24	68.44	68.51	89.51	68.39	10.49	31.49
KNN	84.95	69.97	69.91	89.94	69.59	10.06	30.09

Table 11. Machine Learning Algorithms' accuracy and overall performance.

Model	Accuracy	Precision	Recall	Specificity	F1_score	FPR	FNR
DenseNet121	94.68	89.55	88.3	96.43	88.6	3.57	11.7
ResNet101	96.1	92.03	91.5	97.4	91.66	2.6	8.5
ResNet50	96.22	92.1	92.23	97.51	92.12	2.49	7.77

Table 12. Deep Learning Algorithms' accuracy and overall performance.

5. CONCLUSION AND FUTURE WORKS

The comparison of traditional machine learning algorithms (Decision Tree, Random Forest, KNN) and deep learning models (DenseNet121, ResNet101, ResNet50) for the detection of retinal diseases demonstrates a clear performance advantage for deep learning approaches. While the classical algorithms achieved respectable accuracy levels, peaking at 84.95% for KNN, deep learning models notably outperformed them, with ResNet50 reaching the highest overall accuracy of 96.22%. Moreover, the precision, recall, specificity, and F1 scores were consistently higher for deep learning models, indicating their superior ability to correctly identify disease cases and reduce both false positives and false negatives. In particular, the false positive rate (FPR) and false negative rate (FNR) were substantially lower in ResNet50 (2.49% and 7.77%, respectively) compared to the best-performing classical algorithm, KNN (10.06% and 30.09%).

These results highlight the significant potential of leveraging advanced deep learning architectures for early and reliable detection of retinal diseases, especially in settings with limited access to healthcare professionals and diagnostic infrastructure, such as rural India. By enabling automated and highly accurate diagnosis from retinal images, these models can play a critical role in facilitating large-scale screening programs, allowing for earlier intervention and better disease management. Implementing such solutions could bridge the healthcare accessibility gap, ultimately contributing to the reduction of preventable blindness and improving the overall quality of life for individuals in underserved communities.

Future research should focus on expanding the dataset to include more diverse patient populations and exploring multi-modal data integration to enhance predictive accuracy. This includes combining retinal imaging with medical history, genetic information, and developing hybrid machine learning models that leverage both traditional and deep learning techniques.

Practical implementation will be crucial, involving the development of user-friendly mobile and web applications for remote healthcare settings, conducting clinical validation trials, and creating training programs for healthcare professionals. Collaborative efforts between AI researchers, ophthalmologists, and public health experts will be essential in translating these advanced diagnostic technologies into real-world healthcare interventions.

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